W4111 Exam Study Guide

MongoDB `classicmodels` and Neo4j Movie Database

Professor Guidance

Your professor's guidance for this material is:

> Understand the `classicmodels` sample data from HW3 for MongoDB. Understand the Neo4j Movie Database tutorial from HW3 setup. Be able to query both datasets. For MongoDB, be comfortable with these stages/operators: `$count`, `$group`, `$limit`, `$lookup`, `$match`, `$project`, `$sort`, `$unwind`.

This guide is built around that guidance.

It does three things:

1. Summarizes the two datasets.
2. Reviews the MongoDB operators your professor explicitly listed.
3. Gives varied sample questions with the answer immediately after each question, plus an explanation of what each part of the code does.

There is no separate answer key section.

Part 1 - The Datasets

1.1 MongoDB: `classicmodels`

HW3 loads two collections into MongoDB:

- `classicmodels.customers`
- `classicmodels.orders`

`customers`

Each document is one customer.

```
{
```

```
  "customerNumber": 103,
  "customerName": "Atelier graphique",
  "contact": {
    "lastName": "Schmitt",
    "firstName": "Carine "
  },
  "phone": "40.32.2555",
  "address": {
    "addressLine1": "54, rue Royale",
    "addressLine2": null,
    "city": "Nantes",
    "state": null,
    "country": "France",
    "postalCode": "44000"
  },
  "salesRepNumber": 1370,
  "creditLimit": 21000.0
}
```

Fields to remember:

- `customerNumber`
- `customerName`
- `contact.firstName`, `contact.lastName`
- `phone`
- `address.city`, `address.country`
- `salesRepNumber`
- `creditLimit`

`orders`

Each document is one order. The important difference is that line items are embedded in an array called `orderDetails`.

```
{
  "orderNumber": 10100,
  "orderDate": "2003-01-06",
  "requiredDate": "2003-01-13",
  "shippedDate": "2003-01-10",
  "status": "Shipped",
  "comments": null,
  "customerNumber": 363,
  "orderDetails": [
    {
      "productCode": "S18_1749",
      "quantityOrdered": 30,
      "priceEach": 136.0,
      "orderLineNumber": 3
    }
  ]
}
```

Fields to remember:

- top-level order fields: `orderNumber`, `orderDate`, `status`, `customerNumber`
- embedded array: `orderDetails`
- nested line fields:
- `orderDetails.productCode`
- `orderDetails.quantityOrdered`
- `orderDetails.priceEach`
- `orderDetails.orderLineNumber`

MongoDB exam mental model

- If the question only asks for customer attributes, think `find()` or `$match + $project`.
- If the question wants one row per order line, think `$unwind`.
- If the question wants totals or counts, think `$group`.
- If the question wants data from both `orders` and `customers`, think `$lookup`.

1.2 Neo4j: Movie Database

The HW3 Neo4j problems use the sample Movie graph.

Main node labels:

- `Person`
- `Movie`

Important properties:

- `Person.name`
- `Movie.title`

Important relationships:

- `(:Person)-[:ACTED_IN]->(:Movie)`
- `(:Person)-[:DIRECTED]->(:Movie)`
- `(:Person)-[:WROTE]->(:Movie)`
- `(:Person)-[:PRODUCED]->(:Movie)`
- `(:Person)-[:REVIEWED]->(:Movie)`
- `(:Person)-[:FOLLOWS]->(:Person)`

One especially important detail:

- `ACTED_IN` can have a relationship property `roles`, which is a list.

Neo4j exam mental model

- If the question asks who directed a movie, match `DIRECTED`.
- If the question asks who acted in a movie, match `ACTED_IN`.
- If the question asks who acted with someone, use a shared movie node in the middle.
- If the question asks about a role like `"Neo"`, remember that `roles` is on the relationship, not the node.

Part 2 - MongoDB Operator Cheat Sheet

Your professor explicitly listed these operators. Know what each one does.

`$match`

Filters documents.

```
{ "$match": { "address.country": "France" } }
```

Think SQL `WHERE`.

`$project`

Chooses fields, renames fields, or computes fields.

```
{ "$project": {
    "_id": 0,
    "customerNumber": 1,
    "name": "$customerName",
    "country": "$address.country"
} }
```

Think SQL `SELECT`.

`$sort`

Sorts rows.

```
{ "$sort": { "creditLimit": -1, "customerName": 1 } }
```

- `1` means ascending
- `-1` means descending

`$limit`

Keeps only the first `n` rows.

```
{ "$limit": 5 }
```

Usually appears after `$sort`.

`$unwind`

Flattens an array into one row per array element.

```
{ "$unwind": "$orderDetails" }
```

This is why `orders` questions often start with `$unwind`.

`$group`

Groups rows by a key and computes aggregates.

```
{ "$group": {
    "_id": "$customerNumber",
    "nOrders": { "$sum": 1 }
} }
```

Common patterns:

- count rows: `{ "$sum": 1 }`
- add field values: `{ "$sum": "$someField" }`
- add computed values: `{ "$sum": { "$multiply": [...] } }`

`$count`

Counts how many rows reach this stage.

```
{ "$count": "nOrders" }
```

Outputs one document like:

```
{ "nOrders": 326 }
```

`$lookup`

Joins one collection to another.

```
{ "$lookup": {
    "from": "customers",
    "localField": "customerNumber",
    "foreignField": "customerNumber",
    "as": "cust"
} }
```

Important:

- `$lookup` always creates an array field.
- You often need `$unwind` right after it.

Part 3 - Neo4j Quick Cheat Sheet

Basic pattern

```
MATCH (p:Person)-[:DIRECTED]->(m:Movie)
RETURN p.name AS director, m.title AS movie
ORDER BY director, movie;
```

Aggregation

```
MATCH (p:Person)-[:ACTED_IN]->(m:Movie)
RETURN p.name AS actor, count(m) AS nMovies
ORDER BY nMovies DESC;
```

Filter on relationship property

```
MATCH (p:Person)-[r:ACTED_IN]->(m:Movie)
WHERE 'Neo' IN r.roles
RETURN p.name, m.title;
```

`UNWIND` in Cypher

```
MATCH (p:Person)-[r:ACTED_IN]->(m:Movie)
UNWIND r.roles AS role
RETURN p.name, m.title, role;
```

`WITH`

Use `WITH` to continue after aggregation.

```
MATCH (p:Person)-[:ACTED_IN]->(m:Movie)
```

```
WITH p, count(m) AS nFilms
WHERE nFilms >= 4
RETURN p.name, nFilms;
```

Part 4 - MongoDB Practice Questions

All MongoDB answers appear immediately after each question.

> Assume: ```python from pymongo import MongoClient client = MongoClient(...) db = client["classicmodels"] ```

M1 - Nested filter + projection

Question. Produce a list of customers in the USA whose `creditLimit` is at least `100000`. Return only:

- `customerNumber`
- `customerName`
- `creditLimit`
- `address.country`

Answer.

```
filter_ = {
    "address.country": "USA",
    "creditLimit": {"$gte": 100000}
}

projection = {
    "_id": 0,
    "customerNumber": 1,
    "customerName": 1,
    "creditLimit": 1,
    "address.country": 1
}

result = list(db["customers"].find(filter_, projection))
```

What each part does.

- `find()` is enough because this question only filters and projects fields.
- `"address.country": "USA"` uses dot notation on a nested field.
- `"$gte": 100000` means "greater than or equal to 100000."
- `"_id": 0` removes the default `_id`.
- Fields with `1` are included in the result.

Exam takeaway. Not every MongoDB question needs aggregation.

M2 - Sort + limit

Question. Return the 5 customers with the highest `creditLimit`. Output:

- `customerNumber`
- `customerName`
- `creditLimit`

Sort from highest to lowest.

Answer.

```
pipeline = [
    {"$match": {"creditLimit": {"$gt": 0}}},
    {"$sort": {"creditLimit": -1}},
    {"$limit": 5},
    {"$project": {
        "_id": 0,
        "customerNumber": 1,
        "customerName": 1,
        "creditLimit": 1
    }}
]

result = list(db["customers"].aggregate(pipeline))
```

What each part does.

- `$match` removes useless rows before sorting.
- `$sort` ranks rows by `creditLimit` descending.
- `$limit` keeps only the top 5.
- `$project` trims the output to the requested columns.

Exam takeaway. `top N` questions are usually `$sort` followed by `$limit`.

M3 - One row per line item

Question. Produce one document per line item with:

- `orderNumber`
- `orderDate`
- `productCode`

- `quantityOrdered`
- `priceEach`
- `lineValue`

where `lineValue = quantityOrdered * priceEach`.

Sort by `orderNumber`, then `orderLineNumber`.

Answer.

```
pipeline = [
    {"$unwind": "$orderDetails"},
    {"$project": {
        "_id": 0,
        "orderNumber": 1,
        "orderDate": 1,
        "productCode": "$orderDetails.productCode",
        "quantityOrdered": "$orderDetails.quantityOrdered",
        "priceEach": "$orderDetails.priceEach",
        "orderLineNumber": "$orderDetails.orderLineNumber",
        "lineValue": {
            "$round": [
                {"$multiply": [
                    "$orderDetails.quantityOrdered",
                    "$orderDetails.priceEach"
                ]},
                2
            ]
        }
    }},
    {"$sort": {"orderNumber": 1, "orderLineNumber": 1}}
]

result = list(db["orders"].aggregate(pipeline))
```

What each part does.

- `$unwind` turns each `orderDetails` array element into its own working row.
- `$project` keeps order-level fields and lifts line-item fields to the top level.
- `$multiply` computes one line item's value.
- `$round` formats that computed value to 2 decimals.
- `$sort` returns rows in the requested order.

Exam takeaway. If the question says "for each line item," think `$unwind`.

M4 - Exact M3 order total question

Question. Compute the total value of each order, where each line contributes:

`quantityOrdered * priceEach`

Return:

- `orderNumber`
- `totalValue`

Round to 2 decimals and sort by `orderNumber`.

Answer.

```
pipeline = [
    {"$unwind": "$orderDetails"},
    {"$group": {
        "_id": "$orderNumber",
        "totalValue": {
            "$sum": {
                "$multiply": [
                    "$orderDetails.quantityOrdered",
                    "$orderDetails.priceEach"
                ]
            }
        }
    }},
    {"$project": {
        "_id": 0,
        "orderNumber": "$_id",
        "totalValue": {"$round": ["$totalValue", 2]}
    }},
    {"$sort": {"orderNumber": 1}}
]

result = list(db["orders"].aggregate(pipeline))
```

What each part does.

- `$unwind` exposes each line item.
- `$group` groups all line items for the same `orderNumber`.
- `$multiply` computes one line's dollar value.
- `$sum` adds those line values together.
- `$project` renames `_id` back to `orderNumber`.
- `$sort` orders the final results.

Exam takeaway. This is the core HW3 `M3` pattern.

M5 - Count stage versus grouping count

Question. Answer both parts.

1. How many distinct customers have placed at least one order?
2. How many order documents are in `orders`?

Answer.

```
# Part 1
pipeline_1 = [
    {"$group": {"_id": "$customerNumber"}},
    {"$count": "nCustomersWithOrders"}
]

# Part 2
pipeline_2 = [
    {"$count": "nOrders"}
]
```

What each part does.

- In part 1, `$group` creates one row per distinct `customerNumber`.
- The following `$count` counts those grouped rows.
- In part 2, `$count` directly counts documents in the collection.

Exam takeaway. `$count` the stage is different from `{ "$sum": 1 }` inside `$group`.

M6 - Join orders to customers

Question. For each order, return:

- `orderNumber`
- `orderDate`
- `status`
- `customerName`
- `country`

Sort by `orderDate` descending and keep only the first 10 rows.

Answer.

```
pipeline = [
    {"$lookup": {
        "from": "customers",
        "localField": "customerNumber",
        "foreignField": "customerNumber",
        "as": "cust"
    }},
    {"$unwind": "$cust"},
```

```
    {"$project": {
        "_id": 0,
        "orderNumber": 1,
        "orderDate": 1,
        "status": 1,
        "customerName": "$cust.customerName",
        "country": "$cust.address.country"
    }},
    {"$sort": {"orderDate": -1}},
    {"$limit": 10}
]

result = list(db["orders"].aggregate(pipeline))
```

What each part does.

- `$lookup` joins each order to matching customer documents.
- The join result is stored in an array called `cust`.
- `$unwind` converts that array into a normal embedded document.
- `$project` selects the requested fields.
- `$sort` and `$limit` finish the top-10 request.

Exam takeaway. After `$lookup`, you often need `$unwind`.

M7 - Customers with no orders

Question. List the customers who have never placed an order. Return:

- `customerNumber`
- `customerName`

Answer.

```
pipeline = [
    {"$lookup": {
        "from": "orders",
        "localField": "customerNumber",
        "foreignField": "customerNumber",
        "as": "theirOrders"
    }},
    {"$match": {"theirOrders": {"$size": 0}}},
    {"$project": {
        "_id": 0,
        "customerNumber": 1,
        "customerName": 1
    }},
    {"$sort": {"customerNumber": 1}}
]

result = list(db["customers"].aggregate(pipeline))
```

What each part does.

- `$lookup` attaches an array of matching orders to each customer.
- Customers with no orders get an empty array.
- `$match` keeps only rows where that array has size 0.
- `$project` and `$sort` produce the final output.

Exam takeaway. This is the standard MongoDB anti-join pattern.

M8 - Revenue by country

Question. Compute total revenue by country. Return:

- `country`
- `totalRevenue`

Sort by `totalRevenue` descending.

Answer.

```
pipeline = [
    {"$unwind": "$orderDetails"},
    {"$group": {
        "_id": "$customerNumber",
        "revenue": {
            "$sum": {
                "$multiply": [
                    "$orderDetails.quantityOrdered",
                    "$orderDetails.priceEach"
                ]
            }
        }
    }},
    {"$lookup": {
        "from": "customers",
        "localField": "_id",
        "foreignField": "customerNumber",
        "as": "cust"
    }},
    {"$unwind": "$cust"},
    {"$group": {
        "_id": "$cust.address.country",
        "totalRevenue": {"$sum": "$revenue"}
    }},
    {"$project": {
        "_id": 0,
        "country": "$_id",
        "totalRevenue": {"$round": ["$totalRevenue", 2]}
    }},
    {"$sort": {"totalRevenue": -1}}
]
```

```
result = list(db["orders"].aggregate(pipeline))
```

What each part does.

- First `$unwind` and `$group` compute revenue per customer.
- `$lookup` brings in customer documents.
- Second `$group` changes the level of aggregation from customer to country.
- `$project` formats the grouped result.
- `$sort` ranks countries.

Exam takeaway. It is normal to use more than one `$group` in a single pipeline.

M9 - Broader mixed-operator question

Question. Find the top 3 customers in France by total revenue. Return:

- `customerNumber`
- `customerName`
- `city`
- `totalRevenue`
- `nDistinctProducts`

Answer.

```
pipeline = [
    {"$unwind": "$orderDetails"},
    {"$group": {
        "_id": "$customerNumber",
        "totalRevenue": {
            "$sum": {
                "$multiply": [
                    "$orderDetails.quantityOrdered",
                    "$orderDetails.priceEach"
                ]
            }
        },
        "products": {"$addToSet": "$orderDetails.productCode"}
    }},
    {"$lookup": {
        "from": "customers",
        "localField": "_id",
        "foreignField": "customerNumber",
        "as": "cust"
    }},
    {"$unwind": "$cust"},
    {"$match": {"cust.address.country": "France"}},
    {"$project": {
        "_id": 0,
        "customerNumber": "$_id",
```

```
            "customerName": "$cust.customerName",
            "city": "$cust.address.city",
            "totalRevenue": {"$round": ["$totalRevenue", 2]},
            "nDistinctProducts": {"$size": "$products"}
        }},
        {"$sort": {"totalRevenue": -1}},
        {"$limit": 3}
    ]

    result = list(db["orders"].aggregate(pipeline))
```

What each part does.

- `$unwind` exposes line items.
- First `$group` computes revenue per customer and builds a set of product codes.
- `$lookup` and `$unwind` bring in customer information.
- `$match` keeps only French customers.
- `$project` computes the size of the distinct product set and formats fields.
- `$sort` and `$limit` return the top 3.

Exam takeaway. This kind of question combines several simple ideas into one longer pipeline.

Part 5 - Neo4j Practice Questions

All Neo4j answers appear immediately after each question.

N1 - Directors and movies

Question. Return every director's name and the title of each movie they directed. Sort by director, then movie.

Answer.

```
MATCH (p:Person)-[:DIRECTED]->(m:Movie)
RETURN p.name AS director, m.title AS movie
ORDER BY director, movie;
```

What each part does.

- `MATCH` defines the graph pattern.
- `(p:Person)` matches a person node.
- `-[:DIRECTED]->(m:Movie)` follows a directed edge to a movie.
- `RETURN` selects the properties to output.

- `AS` renames output columns.
- `ORDER BY` sorts the final rows.

Exam takeaway. This is the basic Neo4j traversal pattern.

N2 - Director, movie, actor

Question. Return:

- `director_name`
- `movie_title`
- `actor_name`

for each movie that has both a director and an actor. Sort by `director_name`, then `movie_title`.

Answer.

```
MATCH (d:Person)-[:DIRECTED]->(m:Movie)<-[:ACTED_IN]-(a:Person)
RETURN d.name AS director_name,
       m.title AS movie_title,
       a.name AS actor_name
ORDER BY director_name, movie_title, actor_name;
```

What each part does.

- The movie node `m` is the shared middle node.
- Left side finds the director.
- Right side finds the actor.
- `RETURN` outputs one row per matched combination.

Exam takeaway. This is the same structural idea as HW3 `N2`.

N3 - Co-actor query

Question. Return all people who acted in a movie with `Keanu Reeves`. Exclude `Keanu Reeves` himself. Return:

- `movie_title`
- `coactor`

Sort by `movie_title`, then `coactor`.

Answer.

```
MATCH (k:Person {name: 'Keanu Reeves'})-[:ACTED_IN]->(m:Movie)
      <-[:ACTED_IN]-(co:Person)
WHERE co.name <> 'Keanu Reeves'
RETURN m.title AS movie_title, co.name AS coactor
ORDER BY movie_title, coactor;
```

What each part does.

- `{name: 'Keanu Reeves'}` anchors the query on one specific person.
- First `ACTED_IN` edge finds movies Keanu acted in.
- Second `ACTED_IN` edge finds other actors in those same movies.
- `WHERE` removes the self-match.

Exam takeaway. Co-actor questions almost always have a shared movie node in the middle.

N4 - Count movies per actor

Question. For each actor, return their name and the number of movies they acted in. Only keep actors with at least 4 movies. Sort by count descending, then name ascending.

Answer.

```
MATCH (p:Person)-[:ACTED_IN]->(m:Movie)
WITH p, count(m) AS nFilms
WHERE nFilms >= 4
RETURN p.name AS actor, nFilms
ORDER BY nFilms DESC, actor;
```

What each part does.

- `MATCH` finds all actor-movie pairs.
- `WITH p, count(m) AS nFilms` groups by actor and computes a count.
- `WHERE nFilms >= 4` filters the aggregated rows.
- `RETURN` outputs the final grouped results.

Exam takeaway. `WITH` in Cypher is the normal way to continue after aggregation.

N5 - Role stored on relationship

Question. List each actor who played the role `"Neo"` and the movie where they played it.

Answer.

```
MATCH (p:Person)-[r:ACTED_IN]->(m:Movie)
WHERE 'Neo' IN r.roles
RETURN p.name AS actor, m.title AS movie
ORDER BY actor, movie;
```

What each part does.

- `[r:ACTED_IN]` gives the relationship a variable name.
- `r.roles` accesses the roles list on the relationship.
- `'Neo' IN r.roles` checks whether `"Neo"` appears in that list.

Exam takeaway. Relationship properties are not stored on the nodes.

N6 - `UNWIND` roles into separate rows

Question. For each `ACTED_IN` relationship, produce one row per role. Return:

- `actor`
- `movie`
- `role`

Answer.

```
MATCH (p:Person)-[r:ACTED_IN]->(m:Movie)
UNWIND r.roles AS role
RETURN p.name AS actor, m.title AS movie, role
ORDER BY actor, movie, role;
```

What each part does.

- `r.roles` is a list.
- `UNWIND` expands that list into multiple rows.
- `RETURN` outputs one row per role value.

Exam takeaway. Cypher `UNWIND` is similar in spirit to MongoDB `$unwind`.

N7 - Top 5 directors

Question. Return the 5 directors who have directed the most movies. Return:

- `director`
- `nMovies`

Sort highest to lowest.

Answer.

```
MATCH (p:Person)-[:DIRECTED]->(m:Movie)
RETURN p.name AS director, count(m) AS nMovies
ORDER BY nMovies DESC, director
LIMIT 5;
```

What each part does.

- `MATCH` finds director-movie pairs.
- `count(m)` counts movies per director.
- `ORDER BY` ranks the grouped rows.
- `LIMIT 5` keeps the top 5.

Exam takeaway. This is the Cypher version of group + sort + limit.

N8 - Same person directs and acts

Question. List people who both acted in and directed the same movie.

Answer.

```
MATCH (p:Person)-[:DIRECTED]->(m:Movie)<-[:ACTED_IN]-(p)
RETURN DISTINCT p.name AS name, m.title AS movie
ORDER BY name, movie;
```

What each part does.

- The same variable `p` appears on both sides of the pattern.
- That forces the same person to satisfy both conditions.
- `DISTINCT` removes duplicates.

Exam takeaway. Reusing a variable means "this must be the same node."

N9 - Shortest path

Question. Find the shortest path of any relationships between Tom Hanks and Kevin Bacon.

Answer.

```
MATCH p = shortestPath(
  (t:Person {name: 'Tom Hanks'})-[*]-(k:Person {name: 'Kevin Bacon'})
)
RETURN p;
```

What each part does.

- `shortestPath(...)` asks for the shortest path matching the pattern.
- `[*]` means any relationship type and any direction.
- `p = ...` stores the full path in variable `p`.

Exam takeaway. This is broader than HW3, but it is good practice for reading path syntax.

N10 - Shared-movie count with Tom Hanks

Question. For every actor who has acted with Tom Hanks, return:

- `coactor`
- `sharedMovies`

Return the top 10 by number of shared movies.

Answer.

```
MATCH (tom:Person {name: 'Tom Hanks'})-[:ACTED_IN]->(m:Movie)
      <-[:ACTED_IN]-(co:Person)
WHERE co <> tom
RETURN co.name AS coactor, count(DISTINCT m) AS sharedMovies
ORDER BY sharedMovies DESC, coactor
LIMIT 10;
```

What each part does.

- The pattern finds Tom Hanks, then movies he acted in, then co-actors in those movies.
- `WHERE co <> tom` removes Tom Hanks himself.
- `count(DISTINCT m)` counts different shared movies.
- `ORDER BY` and `LIMIT` return the top 10.

Exam takeaway. This combines the HW3 co-actor idea with aggregation and ranking.

Part 6 - Common Pitfalls and Last Review

Common pitfalls

1. Do not forget that `orderDetails` is an array. Many `orders` questions need `$unwind`.
2. Do not confuse counting rows with summing values:

- count rows: `{ "$sum": 1 }`
- sum field values: `{ "$sum": "$field" }`
- sum computed values: `{ "$sum": { "$multiply": [...] } }`

3. After `$group`, the grouping key is stored in `_id`.
4. After `$lookup`, the joined result is still an array unless you `$unwind` it.
5. In Cypher, relationship direction matters.
6. In Cypher, relationship properties like `roles` require a relationship variable.
7. In Cypher, grouping is implicit whenever you mix aggregates with non-aggregated expressions.

Last-day checklist

- [ ] I can explain the shape of `customers` and `orders`.
- [ ] I know what each of the eight MongoDB operators my professor listed does.
- [ ] I can solve an M3-style total-value question from scratch.
- [ ] I know when a MongoDB question needs `$lookup`.
- [ ] I can write a co-actor query in Neo4j.
- [ ] I can count per person or per movie in Cypher.
- [ ] I can explain what each line of my query is doing instead of only memorizing syntax.

Good luck!

W4111 — Exam Study Guide

MongoDB (`classicmodels`) + Neo4j (sample Movie DB)

How to use this guide

Your professor's exam guidance was explicit:

> *"I **strongly** advise you to understand the `classicmodels` sample data from homework 3 for MongoDB. You are also responsible for having taken the Neo4j tutorial for the Movie Database that was part of homework 3 setup. You are responsible for being able to query the `classicmodels` data and the Neo4j sample movie dataset. I do provide a summary of the data in the exam, but you should ensure you understand it from the homework. WRT to MongoDB, you are responsible for understanding how to use the following aggregation stages/operators: count, group, limit, lookup, match, project, sort, unwind."*

This guide mirrors that guidance exactly:

1. A **data-model refresher** for both datasets — structure, relationships, and the denormalized shape HW3 loaded into MongoDB.
2. A **stage-by-stage MongoDB cheat sheet** covering only the eight operators the professor listed.
3. A **Cypher / Neo4j cheat sheet** matched to the Movie graph you actually worked with.
4. **Practice problems in HW3 style** — solutions provided — that should feel indistinguishable from M1/M2/M3 and N1/N2/N3.
5. A **common-pitfalls** appendix.

If you can work every practice problem in this guide without peeking, you are in good shape.

Part 1 — The `classicmodels` data (MongoDB)

1.1 The original relational schema

`classicmodels` models a small scale-model retailer. In its native SQL form there are eight tables:

Table	Row count	PK	Important FKs
`offices`	7	`officeCode`	—
`employees`	23	`employeeNumber`	`officeCode`, `reportsTo` *(self-join)*
`customers`	122	`customerNumber`	`salesRepEmployeeNumber`
`productlines`	7	`productLine`	—
`products`	110	`productCode`	`productLine`
`orders`	326	`orderNumber`	`customerNumber`
`orderdetails`	2 996	(`orderNumber`, `orderLineNumber`)	`orderNumber`, `productCode`
`payments`	273	(`customerNumber`, `checkNumber`)	`customerNumber`

Core relationship chain (memorize this — every interesting query traverses part of it):

```
productlines --1:N--> products
                        |
                        N:M (via orderdetails line items)
                        |
orders --1:N--> orderdetails
  |
  N:1
  |
customers --N:1--> employees --N:1--> offices
  |
  1:N
```

```
  |
payments
```

1.2 What HW3 actually loaded into MongoDB

HW3 only loads **two collections**, and both are denormalized. This matters because it determines which pipelines even make sense.

`classicmodels.customers` — 122 documents

```
{
  customerNumber: 103,
  customerName: "Atelier graphique",
  contact: { lastName: "Schmitt", firstName: "Carine " },
  phone: "40.32.2555",
  address: {
    addressLine1: "54, rue Royale",
    addressLine2: null,
    city: "Nantes",
    state: null,
    country: "France",
    postalCode: "44000"
  },
  salesRepNumber: 1370,
  creditLimit: 21000.0
}
```

Notes:

- `contact` and `address` are embedded sub-documents. Query nested fields with **dot notation**: `"address.country": "France"`.
- `salesRepNumber` is a foreign key to `employees` — not embedded.

`classicmodels.orders` — 326 documents

```
{
  orderNumber: 10100,
  orderDate: "2003-01-06",          // note: string, not Date
  requiredDate: "2003-01-13",
  shippedDate: "2003-01-10",
  status: "Shipped",
  comments: null,
  customerNumber: 363,              // FK → customers
  orderDetails: [
    { productCode: "S18_1749", quantityOrdered: 30, priceEach: 136.00, orderLineNumber: 3 },
    { productCode: "S18_2248", quantityOrdered: 50, priceEach:  55.09, orderLineNumber: 2 },
    { productCode: "S18_4409", quantityOrdered: 22, priceEach:  75.46, orderLineNumber: 4 },
    { productCode: "S24_3969", quantityOrdered: 49, priceEach:  35.29, orderLineNumber: 1 }
  ]
}
```

Key design facts:

- The relational `orderdetails` table became an **embedded array** inside each `orders` document. That is *the* reason `$unwind` shows up so often.
- There are `2 996` line items total — the number you should see after a single `$unwind "$orderDetails"`.
- Dates are stored as ISO strings like `"2003-01-06"`. Lexicographic sort still yields correct chronological ordering. If you ever need real date math, use `$toDate` first.

Part 2 — MongoDB aggregation cheat sheet

The professor listed exactly these stages/operators: **`$count, $group, $limit, $lookup, $match, $project, $sort, $unwind`**. Learn them cold.

2.1 SQL ↔ MQL mental map

| SQL | MQL stage | Note | |---|---|---| | `WHERE` | `$match` | Put as early as possible (uses indexes). | | `SELECT col1, col2 AS x, expr` | `$project` | 1 = include, 0 = exclude, or assign an expression. | | `ORDER BY` | `$sort` | `{ field: 1 }` asc, `{ field: -1 }` desc. Multi-key preserves order. | | `LIMIT` | `$limit` | Always pair with `$sort` for deterministic output. | | `SELECT COUNT(*)` | `$count: "n"` | Terminal stage; output is a single doc `{n: ...}`. | | *(flatten array)* | `$unwind` | Turns one doc w/ array of length *n* into *n* docs. | | `GROUP BY` + aggregates | `$group` | `_id` is the grouping key; everything else is an accumulator. | | `JOIN` | `$lookup` | Always returns an **array** field; usually followed by `$unwind`. |

2.2 Each stage, with a `classicmodels` example

`$match` — WHERE

```
{ $match: { "address.country": "France" } }
```

Supports full query-operator syntax: `$eq`, `$ne`, `$gt`, `$lt`, `$gte`, `$lte`, `$in`, `$nin`, `$and`, `$or`, `$regex`, `$exists`.

`$project` — SELECT / reshape

```
{ $project: {
    _id: 0,                          // drop _id
    customerNumber: 1,               // keep as-is
    name: "$customerName",           // rename
    country: "$address.country",     // lift nested
    tier: { $cond: [ { $gt: ["$creditLimit", 100000] }, "gold", "silver" ] }
}}
```

Remember: reference a field value with the `$`-prefix (`"$customerName"`), include/exclude with `1`/`0`.

`$sort` — ORDER BY

```
{ $sort: { status: 1, orderDate: -1 } }
```

`$limit` — LIMIT

```
{ $limit: 10 }
```

`$count` — COUNT(*)

```
{ $count: "nOrders" }   // → [ { nOrders: 326 } ]
```

This is the **stage** named `$count`. There is also an accumulator `{ $sum: 1 }` used inside `$group`. Both exist, both are on the syllabus.

`$unwind` — flatten an array

```
{ $unwind: "$orderDetails" }
```

After this stage, each order document exists once per line item, with `orderDetails` now pointing to a single element (not an array). If an array is empty or missing, the document is **dropped** by default — add `{ preserveNullAndEmptyArrays: true }` to emulate a `LEFT JOIN` effect.

`$group` — GROUP BY + aggregates

```
{ $group: {
    _id: "$status",                    // grouping key (use null for "overall")
    nOrders: { $sum: 1 },              // COUNT(*)
    revenue: { $sum: { $multiply: [ "$orderDetails.priceEach",
                                    "$orderDetails.quantityOrdered" ] } }
}}
```

Common accumulators: `$sum`, `$avg`, `$min`, `$max`, `$first`, `$last`, `$push` (list of values), `$addToSet` (distinct list).

Tip: after `$group` the output field that held your key is named `_id`. Use a downstream `$project` to rename it back (`orderNumber: "$_id"`, `_id: 0`).

`$lookup` — JOIN

Equi-join form:

```
{ $lookup: {
    from: "customers",
    localField: "customerNumber",
    foreignField: "customerNumber",
    as: "cust"
}}
```

`cust` is always an **array** of matched docs (even for 1:1 joins). The idiomatic pattern is:

```
{ $lookup: { from: "customers", localField: "customerNumber",
             foreignField: "customerNumber", as: "cust" } },
{ $unwind: "$cust" },               // or preserveNullAndEmptyArrays: true for LEFT JOIN
```

```
{ $project: { _id: 0, orderNumber: 1, customerName: "$cust.customerName" } }
```

2.3 A model pipeline that exercises every required stage

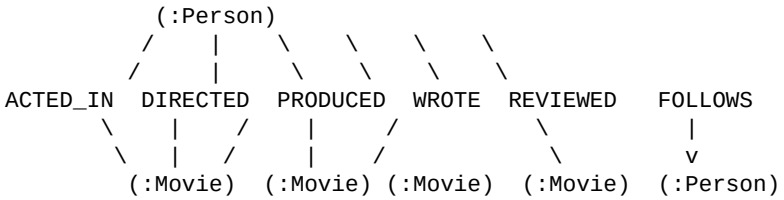
"Top 5 customers by total revenue, with their country and number of orders."

```
db.orders.aggregate([
  { $unwind: "$orderDetails" },
  { $group: {
      _id: "$customerNumber",
      revenue: { $sum: { $multiply: [ "$orderDetails.priceEach",
                                      "$orderDetails.quantityOrdered" ] } },
      nOrders: { $addToSet: "$orderNumber" }
  }},
  { $project: {
      _id: 0,
      customerNumber: "$_id",
      revenue: { $round: ["$revenue", 2] },
      nOrders: { $size: "$nOrders" }
  }},
  { $lookup: { from: "customers", localField: "customerNumber",
               foreignField: "customerNumber", as: "cust" } },
  { $unwind: "$cust" },
  { $match: { "cust.address.country": { $exists: true } } },
  { $project: { customerNumber: 1, revenue: 1, nOrders: 1,
                customerName: "$cust.customerName",
                country: "$cust.address.country" } },
  { $sort: { revenue: -1 } },
  { $limit: 5 }
])
```

That single pipeline uses: `$unwind`, `$group`, `$project`, `$lookup`, `$match`, `$sort`, `$limit`. Add `{ $count: "n" }` to the tail if the question were instead *"how many customers had any revenue?"*.

Part 3 — The Neo4j Movie dataset

3.1 Schema in one picture (ASCII)



Two node labels:

- `(:Movie { title, released, tagline })`
- `(:Person { name, born })`

Six relationship types, **all directed**:

| Rel | From → To | Properties | |---|---|---| | `ACTED_IN` | Person → Movie | `roles: [String]` | | `DIRECTED` | Person → Movie | — | | `PRODUCED` | Person → Movie | — | | `WROTE` | Person → Movie | — | | `REVIEWED` | Person → Movie | `summary`, `rating` | | `FOLLOWS` | Person → Person | — |

Critical gotchas:

- `ACTED_IN.roles` is a **list on the relationship**, not on the node. To get roles use `MATCH (p)-[r:ACTED_IN]->(m) RETURN r.roles`.
- Every movie-related rel points **from Person to Movie** (arrow direction matters for `<-` vs `->` in Cypher patterns).
- `FOLLOWS` is person-to-person (the only rel that isn't movie-bound).

3.2 Cypher cheat sheet

| Concept | Syntax | |---|---| | Node pattern | `(p:Person)` — variable `p`, label `Person` | | Node + property filter | `(p:Person { name: 'Tom Hanks' })` | | Directed rel | `-[:ACTED_IN]->` | | Rel variable | `-[r:ACTED_IN]->` then `r.roles` | | Any-direction / undirected | `-[:FOLLOWS]-` (both ways), or `-[*]-` | | Variable-length | `-[:FOLLOWS*1..3]->` | | Predicate | `WHERE m.released >= 2000 AND m.title CONTAINS 'Matrix'` | | Return | `RETURN DISTINCT p.name AS name, count(m) AS n` | | Sort / page | `ORDER BY n DESC SKIP 10 LIMIT 5` | | Shortest path | `shortestPath((a)-[*]-(b))` | | List ops | `collect(m.title) AS titles`, `size(titles)`, `x IN list`, `[x IN list WHERE ...]` |

Grouping is **implicit**: any non-aggregated variable in `RETURN` acts like `GROUP BY`.

3.3 MongoDB ↔ Cypher one-liner map

| MongoDB | Cypher | |---|---| | `$match` | `WHERE` (or inline pattern filter) | | `$project` | `RETURN ... AS ...` | | `$sort` / `$limit` | `ORDER BY` / `LIMIT` | | `$group` + `$sum/$avg` | implicit grouping + `count()/sum()/avg()` | | `$unwind` | `UNWIND listProperty AS x` | | `$lookup` | walk the relationship: `-[:ACTED_IN]->` | | `$count` | `RETURN count(*)` |

Part 4 — MongoDB practice problems (HW3 style)

Every question below is in the style of M1/M2/M3 in your HW3. Solve first, then verify against the solution.

> **Setup assumed** in all solutions: ```python from pymongo import MongoClient client = MongoClient(...) db = client["classicmodels"] ```

M-Practice 1 — Filter + projection (M1-style)

Prompt. Produce a list of `customers` whose `country` is `"USA"` **and** `creditLimit` is at least `100000`. Your answer should contain only the fields `customerNumber`, `customerName`, `creditLimit`, and `address.country`.

Answer.

```
filter_ = {
    "address.country": "USA",
    "creditLimit": { "$gte": 100000 }
}
projection = {
    "_id": 0,
    "customerNumber": 1,
    "customerName": 1,
    "creditLimit": 1,
    "address.country": 1,
}
result = list(db["customers"].find(filter_, projection))
```

Why it works. `find()` takes a filter (`WHERE`) and a projection (`SELECT`). Dot-notation `address.country` reaches inside the embedded sub-document for both the filter and the projection.

Validation. Expect a small list (roughly a dozen) — all with `country = "USA"` and `creditLimit >= 100000`.

M-Practice 2 — Aggregation with match + sort + limit

Prompt. Return the 5 customers with the **highest** `creditLimit`. Output fields: `customerNumber`, `customerName`, `creditLimit`. Sort descending.

Answer.

```
pipeline = [
    { "$match":  { "creditLimit": { "$gt": 0 } } },
    { "$sort":   { "creditLimit": -1 } },
    { "$limit":  5 },
    { "$project": { "_id": 0,
                    "customerNumber": 1,
                    "customerName": 1,
                    "creditLimit": 1 }},
]
result = list(db["customers"].aggregate(pipeline))
```

Why it works. Exactly the MQL translation of `SELECT ... FROM customers WHERE creditLimit > 0 ORDER BY creditLimit DESC LIMIT 5`.

M-Practice 3 — Unwind + project (M2-style)

Prompt. Produce one document per **line item** with fields `orderNumber`, `orderDate`, `productCode`, `quantityOrdered`, `priceEach`, `lineValue` (= `priceEach * quantityOrdered`). Sort by `orderNumber`, then

`orderLineNumber`.

Answer.

```
pipeline = [
    { "$unwind": "$orderDetails" },
    { "$project": {
        "_id": 0,
        "orderNumber": 1,
        "orderDate": 1,
        "productCode":     "$orderDetails.productCode",
        "quantityOrdered": "$orderDetails.quantityOrdered",
        "priceEach":       "$orderDetails.priceEach",
        "orderLineNumber": "$orderDetails.orderLineNumber",
        "lineValue": {
            "$round": [
                { "$multiply": [ "$orderDetails.priceEach",
                                 "$orderDetails.quantityOrdered" ] },
                2
            ]
        }
    }},
    { "$sort": { "orderNumber": 1, "orderLineNumber": 1 } },
]
result = list(db["orders"].aggregate(pipeline))
assert len(result) == 2996        # sanity check
```

Why it works. `$unwind` flattens the array so each line becomes its own document; `$project` both keeps top-level order fields and pulls the nested line-item fields up; a computed field is built right inside `$project`.

M-Practice 4 — Group + sum (M3-style)

Prompt. For each `status`, compute:

- `nOrders` — the number of orders with that status,
- `totalRevenue` — the sum over all line items in those orders of `priceEach * quantityOrdered`,

rounded to 2 decimals. Sort by `totalRevenue` descending.

Answer.

```
pipeline = [
    { "$unwind": "$orderDetails" },
    { "$group": {
        "_id": "$status",
        "orders": { "$addToSet": "$orderNumber" },
        "totalRevenue": { "$sum": {
            "$multiply": [ "$orderDetails.priceEach",
                           "$orderDetails.quantityOrdered" ]
        }}
    }},
    { "$project": {
        "_id": 0,
        "status":       "$_id",
        "nOrders":      { "$size": "$orders" },
        "totalRevenue": { "$round": [ "$totalRevenue", 2 ] }
```

```
    }},
    { "$sort": { "totalRevenue": -1 } },
]
result = list(db["orders"].aggregate(pipeline))
```

Why it works. Grouping on `$status` gives the per-status bucket. `$addToSet` collects distinct `orderNumber`s so `$size` gives the true order count (a plain `$sum: 1` here would count *line items*, which is wrong). `$sum` over a `$multiply` expression gives per-status revenue.

M-Practice 5 — Group + $count stage (two ways)

Prompt (a). How many distinct customers have placed at least one order? **Prompt (b).** Confirm the number of orders in the `orders` collection.

Answer (a).

```
pipeline = [
    { "$group": { "_id": "$customerNumber" } },
    { "$count": "nCustomersWithOrders" },
]
# → [ { 'nCustomersWithOrders': 98 } ]
```

Answer (b).

```
pipeline = [ { "$count": "nOrders" } ]   # → [ { 'nOrders': 326 } ]
```

Why it works. `$count` is a terminal stage that outputs one document whose single field is named by the argument string. Combined with `$group`, it counts **buckets**. Used on its own, it counts **documents**.

M-Practice 6 — Lookup (equi-join with a collection of documents)

Prompt. For every order, return `orderNumber`, `orderDate`, `status`, and the placing customer's `customerName` and `country`. Sort by `orderDate` descending. Limit to 10.

Answer.

```
pipeline = [
    { "$lookup": {
        "from": "customers",
        "localField": "customerNumber",
        "foreignField": "customerNumber",
        "as": "cust"
    }},
    { "$unwind": "$cust" },
    { "$project": {
        "_id": 0,
        "orderNumber": 1,
        "orderDate": 1,
        "status": 1,
        "customerName": "$cust.customerName",
        "country": "$cust.address.country",
```

```
        }},
        { "$sort":  { "orderDate": -1 } },
        { "$limit": 10 },
    ]
    result = list(db["orders"].aggregate(pipeline))
```

Why it works. `$lookup` writes the matched customer documents into the array field `cust`. `$unwind "$cust"` collapses the array (safe here because each order has exactly one matching customer). Subsequent `$project` lifts the needed fields out with dot notation.

Variation — LEFT JOIN. To keep orders with no matching customer, use `{ $unwind: { path: "$cust", preserveNullAndEmptyArrays: true } }`.

M-Practice 7 — Group across collections (lookup + unwind + group)

Prompt. Compute `totalRevenue` by country. Output `{ country, totalRevenue }`, sorted descending.

Answer.

```
    pipeline = [
        { "$unwind": "$orderDetails" },
        { "$group": {
            "_id": "$customerNumber",
            "rev": { "$sum": { "$multiply": [ "$orderDetails.priceEach",
                                              "$orderDetails.quantityOrdered" ]}}
        }},
        { "$lookup": { "from": "customers",
                       "localField": "_id",
                       "foreignField": "customerNumber",
                       "as": "cust" } },
        { "$unwind": "$cust" },
        { "$group": {
            "_id": "$cust.address.country",
            "totalRevenue": { "$sum": "$rev" }
        }},
        { "$project": { "_id": 0,
                        "country": "$_id",
                        "totalRevenue": { "$round": [ "$totalRevenue", 2 ] } } },
        { "$sort": { "totalRevenue": -1 } },
    ]
    result = list(db["orders"].aggregate(pipeline))
```

Why it works. First group reduces to one doc per customer with their revenue. The `$lookup` attaches each customer's country. A second `$group` re-aggregates by country. Two `$group`s in one pipeline is a perfectly normal idiom.

M-Practice 8 — Customers with no orders (anti-join)

Prompt. List `customerNumber` and `customerName` for customers that have **never** placed an order.

Answer.

```
pipeline = [
    { "$lookup": {
        "from": "orders",
        "localField":   "customerNumber",
        "foreignField": "customerNumber",
        "as": "theirOrders"
    }},
    { "$match":   { "theirOrders": { "$size": 0 } } },
    { "$project": { "_id": 0,
                    "customerNumber": 1,
                    "customerName": 1 }},
    { "$sort":    { "customerNumber": 1 } },
]
result = list(db["customers"].aggregate(pipeline))
```

Why it works. `$lookup` populates `theirOrders` with an array of all their orders (possibly empty). `$match` with `$size: 0` keeps only customers whose array is empty — classic anti-join pattern in MQL.

M-Practice 9 — Top-N within groups (pure pipeline, no joins)

Prompt. For each `status`, find the single order with the largest `totalValue` (sum over its line items). Return `status`, `orderNumber`, `totalValue`. Sort by `totalValue` descending.

Answer.

```
pipeline = [
    { "$unwind": "$orderDetails" },
    { "$group": {
        "_id": { "status": "$status", "orderNumber": "$orderNumber" },
        "totalValue": { "$sum": { "$multiply": [ "$orderDetails.priceEach",
                                                 "$orderDetails.quantityOrdered" ]}}
    }},
    { "$sort":  { "_id.status": 1, "totalValue": -1 } },
    { "$group": {
        "_id":        "$_id.status",
        "orderNumber":{ "$first": "$_id.orderNumber" },
        "totalValue": { "$first": "$totalValue" }
    }},
    { "$project": { "_id": 0,
                    "status": "$_id",
                    "orderNumber": 1,
                    "totalValue": { "$round": [ "$totalValue", 2 ] } } },
    { "$sort": { "totalValue": -1 } },
]
result = list(db["orders"].aggregate(pipeline))
```

Why it works. A compound `_id` `{status, orderNumber}` gives per-order totals within status. Sorting then re-grouping on `status` with `$first` picks the max row per group (this is MQL's standard "argmax per group" idiom).

M-Practice 10 — Full kitchen-sink pipeline

Prompt. "Top 3 customers in France by total revenue. Return customer number, customer name, city, total revenue, and the number of distinct products they have ever ordered."

Answer.

```
pipeline = [
    { "$unwind": "$orderDetails" },
    { "$group": {
        "_id": "$customerNumber",
        "rev":      { "$sum": { "$multiply": [ "$orderDetails.priceEach",
                                               "$orderDetails.quantityOrdered" ]}},
        "products": { "$addToSet": "$orderDetails.productCode" }
    }},
    { "$lookup": { "from": "customers",
                   "localField": "_id",
                   "foreignField": "customerNumber",
                   "as": "cust" } },
    { "$unwind": "$cust" },
    { "$match": { "cust.address.country": "France" } },
    { "$project": {
        "_id": 0,
        "customerNumber": "$_id",
        "customerName":   "$cust.customerName",
        "city":           "$cust.address.city",
        "totalRevenue":   { "$round": [ "$rev", 2 ] },
        "nDistinctProducts": { "$size": "$products" },
    }},
    { "$sort":  { "totalRevenue": -1 } },
    { "$limit": 3 },
]
result = list(db["orders"].aggregate(pipeline))
```

Uses every required stage: `$unwind`, `$group`, `$lookup`, `$match`, `$project`, `$sort`, `$limit`. If this came up in the exam, changing it to "how many French customers have ever ordered a 'Classic Cars' product?" is just swapping a `$match` condition and replacing the tail with `{ $count: "n" }`.

Part 5 — Neo4j practice problems (HW3 style)

All queries assume the sample Movie graph loaded by the `:play movies` tutorial.

N-Practice 1 — Single pattern traversal (N1-style)

Prompt. Return every actor's name and the title of each movie they acted in. Sort by actor then movie.

Answer.

```
MATCH (p:Person)-[:ACTED_IN]->(m:Movie)
RETURN p.name AS actor, m.title AS movie
ORDER BY actor, movie;
```

Why it works. This is the acting equivalent of N1. Mirror the N1 pattern, just swap `DIRECTED` → `ACTED_IN`.

N-Practice 2 — Triangle pattern (N2-style)

Prompt. Compute `director_name`, `movie_title`, `writer_name` for every movie that has both a director and a writer. Sort by `director_name`, then `movie_title`.

Answer.

```
MATCH (d:Person)-[:DIRECTED]->(m:Movie)<-[:WROTE]-(w:Person)
RETURN d.name AS director_name,
       m.title AS movie_title,
       w.name AS writer_name
ORDER BY director_name, movie_title;
```

Why it works. Two relationships meeting at the same `(m:Movie)` node — exactly the director/actor pattern from N2 but with `WROTE` instead of `ACTED_IN`.

N-Practice 3 — Co-actor pattern (N3-style)

Prompt. Return the distinct names of all people who acted in a movie with **Keanu Reeves**, ordered by movie title.

Answer.

```
MATCH (k:Person {name: 'Keanu Reeves'})-[:ACTED_IN]->(m:Movie)
      <-[:ACTED_IN]-(co:Person)
WHERE co.name <> 'Keanu Reeves'
RETURN DISTINCT m.title AS movie_title, co.name AS coactor
ORDER BY movie_title, coactor;
```

Why it works. Same "friends-of-friends" pattern as N3; the `WHERE co.name <> ...` filters out the starting node, which otherwise trivially co-acts with itself.

N-Practice 4 — Aggregation (count per group)

Prompt. For each actor, return their name and the number of movies they have acted in. Only return actors with at least 4 movies. Order by count descending, then by name.

Answer.

```
MATCH (p:Person)-[:ACTED_IN]->(m:Movie)
WITH p, count(m) AS nFilms
WHERE nFilms >= 4
RETURN p.name AS actor, nFilms
ORDER BY nFilms DESC, actor;
```

Why it works. Cypher has implicit grouping: the non-aggregated `p` defines the group. `WITH` is Cypher's way to chain subsequent clauses onto the aggregation result — think of it as the subquery break. Compare to MQL's `$group` + `$match`.

N-Practice 5 — Filter by relationship property

Prompt. List each actor who played the role "Neo", and the movie where they played it.

Answer.

```
MATCH (p:Person)-[r:ACTED_IN]->(m:Movie)
WHERE 'Neo' IN r.roles
RETURN p.name AS actor, m.title AS movie;
```

Why it works. Binding the relationship to variable `r` lets you access the `roles` list that lives on the edge. `IN` is Cypher's list-membership check.

N-Practice 6 — Unwind for per-role rows

Prompt. For each `ACTED_IN` relationship, produce one row per role. Output: `actor`, `movie`, `role`.

Answer.

```
MATCH (p:Person)-[r:ACTED_IN]->(m:Movie)
UNWIND r.roles AS role
RETURN p.name AS actor, m.title AS movie, role
ORDER BY actor, movie, role;
```

Why it works. `UNWIND` is Cypher's `$unwind` — it flattens a list property into one row per element.

N-Practice 7 — Top-N with limit

Prompt. Return the 5 directors who have directed the most movies, with counts, ordered high to low.

Answer.

```
MATCH (d:Person)-[:DIRECTED]->(m:Movie)
RETURN d.name AS director, count(m) AS nMovies
ORDER BY nMovies DESC, director
LIMIT 5;
```

N-Practice 8 — Subquery / intersection

Prompt. List people who both **acted in** and **directed** the same movie.

Answer.

```
MATCH (p:Person)-[:DIRECTED]->(m:Movie)<-[:ACTED_IN]-(p)
RETURN DISTINCT p.name AS name, m.title AS movie
ORDER BY name, movie;
```

Why it works. Using the same variable `p` on both ends of the pattern forces the director and actor nodes to be the same person.

N-Practice 9 — Variable-length path (Bacon number)

Prompt. Find the shortest path (of any relationships) between Tom Hanks and Kevin Bacon. Return the path.

Answer.

```
MATCH p = shortestPath(
  (t:Person {name: 'Tom Hanks'})-[*]-(k:Person {name: 'Kevin Bacon'})
)
RETURN p;
```

If the exam instead asks for the **length**: `RETURN length(p) AS baconNumber`.

N-Practice 10 — Multi-hop traversal with aggregation

Prompt. For every actor who has acted with Tom Hanks, list their name and the number of movies they share with him. Return top 10.

Answer.

```
MATCH (tom:Person {name: 'Tom Hanks'})-[:ACTED_IN]->(m:Movie)
      <-[:ACTED_IN]-(co:Person)
WHERE co <> tom
RETURN co.name AS coactor, count(DISTINCT m) AS sharedMovies
ORDER BY sharedMovies DESC, coactor
LIMIT 10;
```

Why it works. Same co-actor pattern as N3 plus grouping via implicit `GROUP BY co` and `count(DISTINCT m)` to avoid double-counting if the same co-actor appeared in the same movie twice.

Part 6 — Common pitfalls & exam tips

1. **MQL aggregation order matters for performance and correctness.** Place `$match` as early as possible. Don't `$unwind` before `$match` if the match can be done on the parent document — you'll create extra work.

2. **After `$lookup`, always decide: `$unwind` (inner-join) or `preserveNullAndEmptyArrays` (left-join)?** Forgetting this is the #1 cause of "why is my result an array?" bugs.
3. **`$group._id` vs `$project`.** After a `$group`, your key is in `_id`. You almost always want a `$project` afterwards to rename it and hide `_id`.
4. **`$addToSet` vs `$push`.** Use `$addToSet` when counting distinct values; `$push` is the SQL analog of `ARRAY_AGG` (keeps dupes).
5. **`$count` the stage ≠ `$count` in a group.** The **stage** is a terminal counter that outputs `{field: n}`. Inside `$group`, you count with `{ $sum: 1 }`.
6. **Strings vs numbers in `classicmodels`.** `orderDate` is a **string**. Sort works because of the ISO format; arithmetic requires `$toDate`.
7. **Cypher direction is mandatory**. `-[:DIRECTED]->` and `<-[:DIRECTED]-` give different results. Undirected is `-[:DIRECTED]-`.
8. **Grouping in Cypher is implicit.** Every non-aggregated variable in `RETURN` is a grouping key. Want to aggregate on `actor` alone? Only mention `actor` (and aggregates) in the `RETURN`.
9. **Relationship properties live on the edge.** To filter by `ACTED_IN.roles`, give the relationship a name: `-[r:ACTED_IN]->` then `WHERE 'Neo' IN r.roles`.
10. **The "I do provide a summary" clause.** On the exam the professor will paste a schema blurb. Don't ignore it — it's the only source of truth for what collections / labels exist in the version you're being tested on.

Quick last-day review checklist

- [ ] I can write in MQL: any HW3-style question using only `$match, $project, $sort, $limit, $unwind, $group, $lookup, $count`.
- [ ] I can compute total order value via `$unwind + $group + $sum + $multiply`.
- [ ] I can do an **anti-join** ("customers without orders") using `$lookup` + `$match: {arr: {$size: 0}}`.
- [ ] I know what the default `$unwind` does to empty arrays and how to change it.
- [ ] I can write `MATCH (a)-[:REL]->(b)<-[:REL]-(c)` triangles and understand co-actor style queries.
- [ ] I can access relationship properties: `-[r:ACTED_IN]-> ... r.roles`.
- [ ] I can count movies per person with implicit grouping.
- [ ] I know how to pair `WITH` with a filter to emulate `HAVING`.

Good luck!